# DOD Retrieval from MTG Satellite Data with Machine Learning (DUSTR)

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## Goal

**Retrieve information on dust from GEO satellite images**

## Background

- Use Meteosat Third Generation (MTG) high-temporal resolution satellite data
- Do retrieval of DOD (Dust Optical Depth)
- There are already analytical methods (have limitations)

## Main research Questions

- Can ML models improve DOD retrieval accuracy?
- Can ML models achieve a generalized retrieval scheme?
- Which are the most important input variables?

# General implementation/workflow

1. Get/create relevant data
2. Train & optimize multiple ML
3. Select/find what works well
4. Analyze scope of ML application
5. Independent validation
6. Final retrieval product/method

# Data

## Target Data Sources

- **SLSTR Level 2 Aerosol Optical Depth - Sentinel-3**
  - $9.5 \times 9.5$ km$^2$
  - Continuous LEO swaths
  - Initial development, now stopped
- **EarthCARE (current use)**
  - Point measurements
  - Continuous LEO swaths
  - OREO product...
    1. **DOD @ 355 nm** from vertical profiles
    2. **Total mass** from vertical profiles

# Data

## Features Data Sources

- **FCI** Level 1c Normal Resolution Image Data - MTG - 0 degree
  - 16 spectral channels / 10 minutes @ 1 - 2 km
- **Cloud Mask** - MTG - 0 degree (from FCI)
  - / 10 minutes @ 2 km
- **ERA5** hourly data on single levels
  - / 1 h @ $0.25° \times 0.25°$
  - 27 variables on the surface
- LSA/SAF MTG Land Surface Temperature (**MTLST**) from FCI
  - / 10 minutes @ 2 km
  - not used for now

## Other Data

- **SZA** calculated for each grid point/time
- **Land mask** created from OSM polygons data

# Data preprocessing

- Date range: **2025-03-19 – 2025-05-10**
- Temporal resolution: **30 minutes**
  - May half temporal resolution and double date range
- Domain:

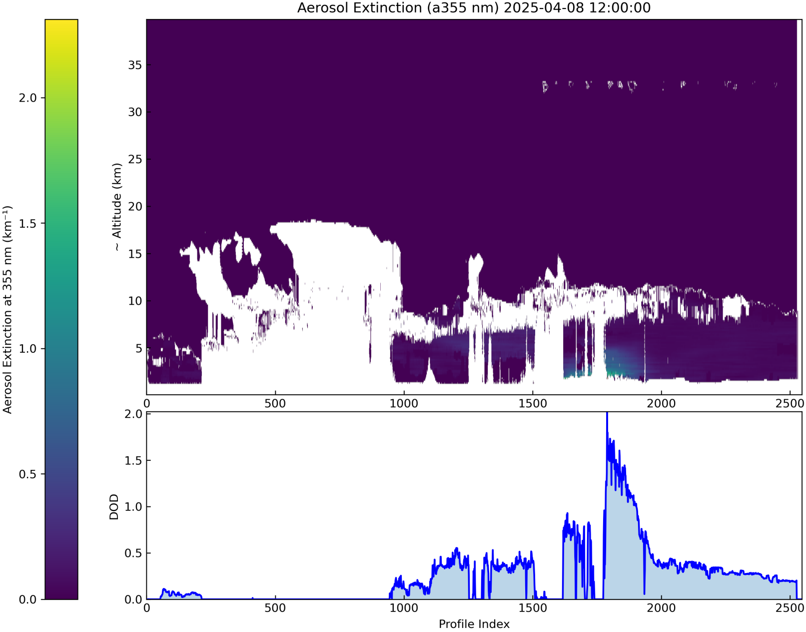
| | | |
|---|---|---|
| | **45N** | |
| **-30W** | | **45E** |
| | **0S** | |

- Regrid on regular grid: $\mathbf{0.02° \times 0.02°}$
  - Near the resolution finer available data
  - Data representation on the grid coordinates (center of cell)

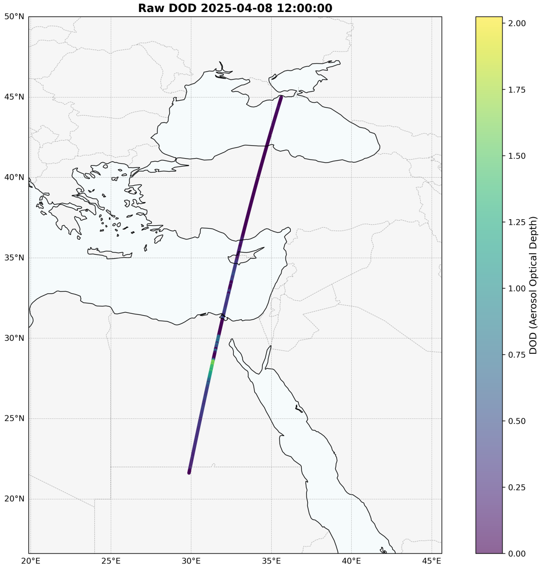
Regrid methods:

- EarthCARE -> mean values within the cells
- Rest of data -> **nearest** values to grid
  - Keep actual values – no spatial smoothing/interpolation
- ERA5 -> temporal interpolation within hourly data

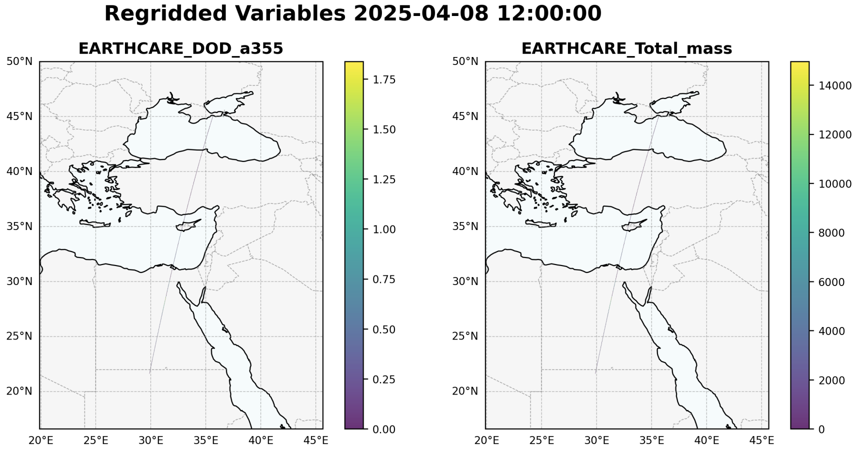
# Raw Data view



# Calculated DOD



# Regridded


Regridded Variables 2025-04-08 12:00:00
EARTHCARE_DOD_a355
EARTHCARE_Total_mass
50°N
45°N
40°N
35°N
30°N
25°N
20°N
20°E
25°E
30°E
35°E
40°E
45°E
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0
2000
4000
6000
8000
10000
12000
14000

# Machine Learning

## Current “easy” approach, well selected data

- Land only
- No clouds
- Sun lit domain

| Dataset | N | Split |
|---|---|---|
| Train | 194442 | 70.1% |
| Valid | 41340 | 14.9% |
| Test | 41550 | 14.9% |

## Potencial problems

- Good results, but maybe too easy too fast ...?
- Apply ML model:
  - over the oceans?
  - with mixed clouds?
  - longer time ranges of other periods of year?

# Model training/tuning iterative process

- Multiple feature importance metrics implemented
- Initial evaluation of 9 ML models
- Keeping most promising:
    - **Decision Tree**
    - **Random Forest**
    - **XGBoost**
- Iterative Hyperparameter optimization (2 approaches)
    - Minimize **MSE** with manual features selection (multiple validation metrics are evaluated)
    - Minimize **MSE** with automatic methods of features selection (multiple validation metrics are evaluated)
    - Multiple validation metrics are evaluated on each batch's best

# Preliminary results

## Multiple ML models with very good metrics!?

| Target | Model | Trials | Best_r2 | Best_rmse | Best_mae |
|---|---|---|---|---|---|
| DOD_a355 | Decision Tree | 673 | 0.9659 | 0.0381 | 0.0171 |
| | Random Forest | 684 | 0.9714 | 0.0349 | 0.0169 |
| | XGBoost | 626 | 0.9749 | 0.0327 | 0.0171 |
| Total_mass | Decision Tree | 750 | 0.9538 | 346.6828 | 137.7835 |
| | Random Forest | 750 | 0.9714 | 272.8909 | 123.6965 |
| | XGBoost | 720 | 0.9762 | 248.7359 | 123.7431 |

## Do you need more metrics?? Propose!

**$R^2$**, **RMSE**, **MAE**, **MSE**, **N_samples**, **pred_min**, **pred_max**, **pred_mean**, **pred_std**

# Things missing

- Cross-validation with independent data
- Multi season evaluation/training
- Regional Validation
  - Mediterranean
  - Middle East
  - Atlantic
  - North Africa
- Analyses of the effects of:
  - Data selection (no cloudy, over land)
  - Features (physical meaning, relation to analytical methods
  - Hyperparameters of the models
- More/other variables

# Thank You!

**Contact:**

- Email: anatsis@noa.gr
- Project: github.com/NOA-ReACT/DUSTR

**Presentation:** github.com/thanasisn/presentations

**Q&A**

# Example of features importance selection

# List of all variables